

Contrastive learning-based few-shot relation extraction with open-book datastore

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ABSTRACT

Few-shot relation extraction (FSRE) aims to identify relations with the aid of few instances. In real-world application scenarios, users often encounter new entities or relations, and there is insufficient labeled data to train the model, making FSRE particularly important. However, limited by the quality of instances in real-world settings, FSRE struggles to discern the subtle semantic differences among instances and corresponding relations. This leads to the critical problem of effectively learning these semantic differences of similar instances and similar relations effectively. In this paper, we propose a novel Few-Shot Contrastive Learning (FSCL) framework to alleviate the semantics challenges of similar instances and relations. To learn the semantic information in low-resource environments, FSCL jointly utilizes contrastive learning with an open-book datastore, while updating the instance representations in the open-book datastore asynchronously. To further enhance the robustness and reduce the impact of semantic bias caused by contrastive learning, we explore six data augmentation strategies on three public datasets to capture more meaningful semantic distinctions among instances as well as their relations. By conducting data augmentation experiments, we find that dropout noise is an effective strategy for data augmentation that can significantly highlight semantic differences between similar instances when learning similar relational representations. This helps the model obtain more potent and meaningful representations. Experiments on five public datasets demonstrate the effectiveness of the proposed method. Our codes are available on <https://github.com/JohnnyGWY/FSCL>.

1. Introduction

Few-shot relation extraction (FSRE) is a crucial task in natural language processing that aims to identify and extract relations between entities from a small amount of sentences. It is essential for developing applications like information retrieval [1], recommender system, knowledge graph construction [2], and intelligent question and answer system [3].

In recent years, the rise of pre-trained language models (PLMs, e.g., GPT, BERT, etc.) has brought new breakthroughs in FSRE. A PLM is employed as a feature extractor and fine-tuned on FSRE [4–6] and zero-shot relation extraction [7,8]. This improves the performance of relation extraction by utilizing the generalized representation capabilities learned by the PLMs on large-scale data and migrating it to the few-shot task. However, despite these advancements, FSRE still faces significant challenges. One major issue is the anisotropy of token distribution after encoding using a PLM on few-shot instances, which results in poor quality instances. The poor quality of instances will result in the model learning incorrect semantic information for similar

instances, leading to insufficient semantic comprehension of similar relations and consequently, poor extraction performance.

Current methods often fail to address this issue effectively. They struggle to distinguish subtle semantic differences among instances (similar instances) that belong to similar relations. For example, we select three similar relation labels from the dataset, which are “Component-Whole” (indicating the relationship between a component and the whole to which it belongs), “Content-Container” (indicating a relationship where one content is contained or stored in a container), and “Member-Collection” (indicating that an individual is a member of a collection or group). Upon careful analysis of the descriptions of these three relations, they all encompass the concept of one entity being part of another entity. Semantically, the three relation labels are quite similar. Consequently, when faced with instances that are similarly semantically too, the model’s prediction becomes challenging. From the examples in Fig. 1, we observe that for instance 1, the gold relation was identified as “Member-Collection” instead of the correct “Content-Container”, and for instance 2, it was identified as “Content-Container” instead of the correct “Component-Whole”.

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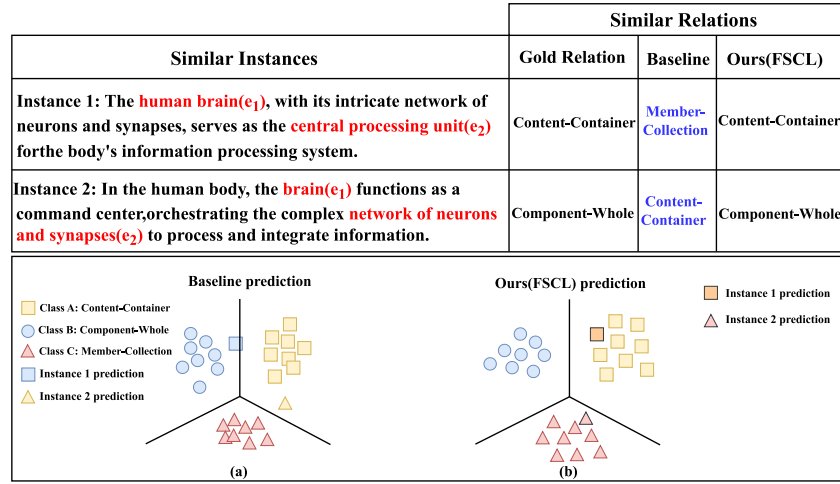


Fig. 1. Examples of relation extraction performance that differences between baseline and ours(FSCL). We conduct predictions using both the baseline method RETRIEVALRE [4] and our proposed FSCL (b). At the top of the picture, we show an example of a pair of instances that struggles to discern the subtle semantic differences among instances and corresponding relations. Instances with entities and incorrect relation extraction results are highlighted in red and blue, respectively. At the bottom of the picture, we select eight instances from each of the three classes within the open-book datastore, the instances within each class exhibit uneven distribution, i.e., similar instances. This phenomenon results in mispredictions of similar relations by the baseline, yielding inaccurate results.

To bridge this gap, our work focuses on improving the quality of instance representations and enhancing the model's ability to learn and differentiate subtle semantic differences. Our central hypothesis is that the subtle semantic differences between similar relations can be effectively captured by enhancing the quality of instance representations through Contrastive Learning (CL). This is motivated by the observation that in high-dimensional representation spaces, instances that are semantically similar should be closer to each other, while those that are dissimilar should be farther apart. CL has been widely used in computer vision [9] to improve the quality of instance representations. In our approach, we extend CL to the few-shot setting by applying it to instances and relations that are semantically similar. The goal is to create a more informative and discriminative feature space where similar instances are well-separated from each other, even in low-resource scenarios. Inspired by Instance-wise CL [9], we utilize the thought of Instance-wise CL on the similar instances and similar relations to make the instances more informative in terms of representation space, that is, it improves the quality of the instances. As shown in Fig. 1(b), after utilizing contrastive learning to the training set, we achieve a balanced distribution of instances in the representation space. Subsequently, during few-shot learning, we obtain instances with more quality successfully.

In this paper, we propose a novel Few-Shot Contrastive Learning (FSCL) framework to deal with the semantic differences of similar instances and similar relations effectively. We reshape the instance representations generated by the PLMs after template processing, in addition to using the retrieval method to effectively solve the problem of anisotropy in the representation space of the instance, as illustrated in Fig. 1(a). We achieve this by drawing an instance and its augmented view closer together in the representation space, while simultaneously pushing away the distance between different sentences and their augmented views. Moreover, we explore different data augmentations strategies to get instance augmented views, and then these views are used to learn semantic differences for similar instances and similar relations. For training stage, we combine contrastive learning loss and relation prediction loss for learning, and for final prediction we combine relation prediction and retrieval modules to extract relations.

To summarize, the main contributions of our work are as follows:

- We propose a novel framework based on Instance-wise contrastive learning for few-shot relation extraction. It jointly optimizes the contrastive learning loss and relation prediction loss in combination with an open-book datastore to extract relations, which

alleviates the semantics problems of similar instances and similar relations effectively by improving the quality of instances.

- We explore six data augmentation strategies to reduce the impact of semantic bias caused by contrastive learning, and the experimental results on three public datasets show that the data augmentation strategy with dropout helps instances to obtain more distinctive representations with more subtle differences between instances.
- We conduct a series of empirical studies on five public datasets. Experimental results show that compared with the state-of-the-art methods FSCL improves the performance of relation extraction and enhances the ability to memorize long-tailed or hard patterns in the case of low-resource environments.

2. Related work

Few-shot Relation Extraction. Few-shot relation extraction (FSRE) is a branch of relation extraction that includes meta-learning [10], graph-based [11,12], and large language models. Zhang et al. [13] combine generic and domain-specific knowledge graphs to improve the global semantics of relation types and performance. Han et al. and Gao et al. [14,15] learn a metric function to find the prototype center of a category in a metric space with a small number of few-shot instances. The testing step uses a nearest-neighbor classifier. These methods require a good initial state or prior knowledge for iterative optimization. Qu et al. [10] introduce external knowledge from Wikidata¹ and construct a global relation graph, optimizing the prior distribution of few-shot relation prototype vectors through different methods. Li et al. [12] train a graph-based model and fine-tune task-specific models on different few-shot datasets while keeping the graph-based model unchanged. These methods involve extracting semantic relations between entities from text and constructing corresponding graph structures, which can be affected by text noise, ambiguity, and complexity, resulting in incomplete or inaccurate graph structures.

In recent years, with the emergence of PLMs, it has become a trend to utilize the powerful semantic representation and the ability to capture long-distance information of PLMs in few-shot relation extraction. Wu et al. [16] enhance PLMs by adding entities with symbols' token

¹ <https://www.wikidata.org/>

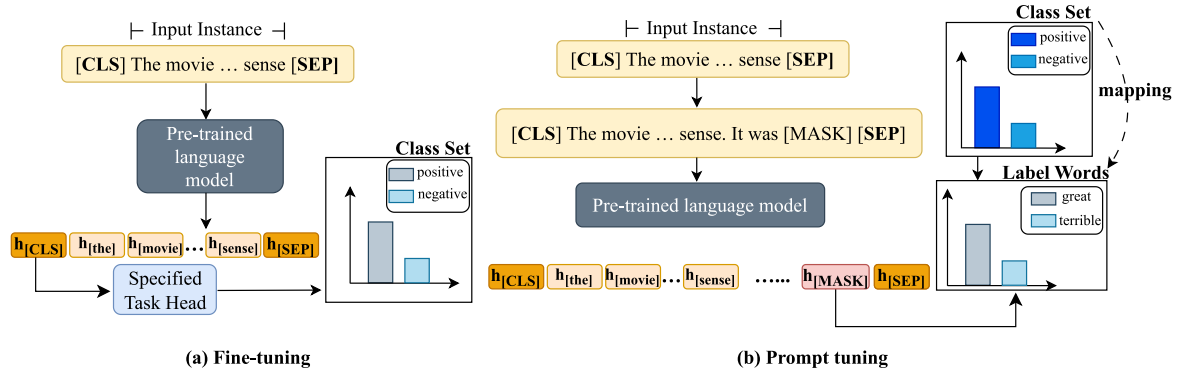


Fig. 2. The differences of fine-tuning and prompt tuning.

and using the outputs to represent the hidden state of the entities, obtaining relation information between entities. Chen et al. [17] consider the richness of previously unnoticed semantic and prior knowledge that exists between relational labels, and they integrated this knowledge into the structure of relation extraction to increase the semantic understanding of the text by PLMs. SPANBERT [18], KNOWBERT [19], and LUKE [20], which utilize external knowledge to improve performance. In contrast, we start from the existing instance itself to mine more information about the instance and also consider the low-resource case.

Prompt-tuning and Retrieval. GPT-3 has introduced a new paradigm, called prompt-tuning, which bridges the gap between Pre-training and Fine-tuning. Wei et al. [21] have applied prompt-tuning to generate a chain of thought, which has significantly improved the inference ability of large language models. Fan et al. [22] design a loss function to align the virtual word prompts of multi-perspective relation representations. Another form of prompt, called sub-prompt, has been created based on prior knowledge and used for classification tasks [23]. The starting point of these models is to make the semantics of sentences comprehensible to a larger model of the language, rather than to refine the semantics of the sentences themselves. The framework we propose, further refines the semantics of the sentences themselves from the starting point.

Prompt's discrete nature [24,25] inspires us to construct a knowledge base that retrieves a small number of instances in a task. Retrieval is often seen as a way to evaluate knowledge rather than contributing to learning. By combining retrieval with prompt-tuning, researches have shown that [26,27] retrieval enhances learning, and incorporating it into learning is a powerful approach.

Contrastive Learning. Contrastive learning was proposed and applied in the field of computer vision [9] to solve unsupervised visual representations. In recent years, contrast learning has also been applied to natural language processing. Kim et al. [28] employ the bert intermediate layer's hidden representation for self-guided contrastive learning, dynamically integrating the knowledge stored in the bert layer to build sentence embeddings. Carlsson et al. [29] apply the final layer representations of the two independent models with noise contrastive learning to counteract the task bias caused by the accumulation of network layers as they are added up, thus making the final layer representations suitable for feature extraction. Giorgi et al. [30] and Wu et al. [31] adopt the same SimCLR architecture and objectives, which often require large corpora and are not suitable for learning with few-shot in low-resource situations.

3. Preliminaries

This section introduces the preliminaries on fine-tuning and prompt tuning related to this work. The important notations are summarized in Table 1. An RE task dataset can be represented as $D = \{\mathcal{X}, \mathcal{Y}\}$, where $\mathcal{X}$ denotes the set of input instances, and $\mathcal{Y}$ represents the set of relation classes. Each i th instance $x^i \in \mathcal{X}$ is composed of multiple tokens,

$x^i = \{w_1^i, w_2^i, \dots, w_k^i, \dots, w_l^i, \dots, w_m^i, \dots, w_n^i, \dots, w_L^i\}$ and corresponds to a class $y^i \in \mathcal{Y}$, L is the length of the original instance. The objective of the RE task is to extract relation(class y) between subject and object entities. To account for the multi-token nature of entities, we employ w_{sub}^i and w_{obj}^i to represent the subject entity $\langle w_k^i, \dots, w_l^i \rangle$ and object entity $\langle w_m^i, \dots, w_n^i \rangle$, respectively.

Fig. 2(a) presents fine-tuning. For a downstream classification task, when working with a pre-trained language model $\mathcal{M}$, the standard fine-tuning procedure involves the initial transformation of the input instance $x^i = \{w_1^i, w_2^i, \dots, w_{sub}^i, \dots, w_{obj}^i, \dots, w_L^i\}$ into a sequence that is formally structured with the insertion of [CLS] at the beginning and [SEP] in the end. [CLS] is an important token used for downstream tasks, while [SEP] is used as a separator between different sequences. Subsequently, $\mathcal{M}$ is employed to encode sequence, generating hidden vector $\mathbf{h} = \{\mathbf{h}_{[CLS]}, \mathbf{h}_{w_1^i}, \mathbf{h}_{w_2^i}, \dots, \mathbf{h}_{w_{sub}^i}, \dots, \mathbf{h}_{w_{obj}^i}, \dots, \mathbf{h}_{[SEP]}\}$. The probability distribution over the class set $\mathcal{Y}$ is then obtained by training a specified task head, where widely used $p(\cdot|x) = \text{Softmax}(\mathbf{W}\mathbf{h}_{[CLS]})$. Here, $\mathbf{W}$ represents a parameter matrix initialized with random values, and $\mathbf{h}_{[CLS]}$ signifies the representation corresponding to [CLS]. The optimization process involves minimizing the cross-entropy loss over $p(\cdot|x)$ concerning X , while optimizing both $\mathbf{W}$ and $\mathcal{M}$.

As shown in Fig. 2(b), standard prompt is made up of a template $\mathcal{T}(\cdot)$ and a label words set $\mathcal{V}$. Specifically, the input instances are mapped into fill-in-the-blank phrases through a template, the output will be denoted as $\hat{x}$, e.g., " $\hat{x} = \mathcal{T}(x) = \langle \text{input instance} \rangle$ It is [MASK].". The set of relation classes $\mathcal{Y}$ is mapped the internal mapping function ϕ (also known as verbalizer) to the label words set $\mathcal{V}$, e.g., "positive" (one class in the set of classes) $\rightarrow$ *great* (one label word in the set of label words), "negative" (one class in the set of classes) $\rightarrow$ *terrible* (one label word in the set of label words). The probability distribution over $\mathcal{Y}$ can be formalized using the verbalizer to calculate the probability distribution over $\mathcal{V}$ at the masked position, i.e. $P(y|x) = P_{\mathcal{M}}([\text{MASK}] = v | \mathcal{T}(x))$, where $v \in \mathcal{V}$, $v = \phi(y)$ and $y \in \mathcal{Y}$, $\mathcal{M}$ is PLM.

4. Approach

4.1. Proposed model overview

As illustrated in Fig. 3, Our framework consists of five steps: Step 1. During the training phase, FSCL processes a batch of few-shot instances, each comprising a pair of head and tail entities. Subsequently, each instance undergoes the input instance encoder, generating contextual instance embeddings denoted as $f(x)$ and prompt-based embeddings h_x . K-nearest neighbors $\mathcal{N}$ of h_x are also retrieved from each class of the training set and form knowledge pairs with the class set, which construct an open-book datastore. Step 2. Taking the contextual instance embeddings $f(x)$ obtained from each batch as input, different relation augmentation layers were designed to obtain the positive instance pairs $f(x^+)$ and the negative instance sets $f(x^-)$ within the batch. Step 3.

Table 1
Important notations.

Notations	Descriptions
$\mathcal{D}$	The dataset of RE task consists of $\mathcal{X}$ and $\mathcal{Y}$.
$\mathcal{X}$	The set of input instances.
$\mathcal{Y}$	The set of relation classes.
x^i	An i th instance in $\mathcal{X}$ and composed of multiple tokens like $x^i = \{w_1^i, w_2^i, \dots, w_L^i\}$.
y^i	The class of instance x^i in $\mathcal{Y}$.
w_{sub}^i	The acronyms of the subject entity $\langle w_k^i, \dots, w_l^i \rangle$.
w_{obj}^i	The acronyms of the object entity $\langle w_m^i, \dots, w_n^i \rangle$.
$\mathcal{M}$	A pre-trained language model.
$[\text{CLS}], [\text{SEP}], [\text{MASK}], [\text{E}]$	Tokens with special processing used in instance.
$\mathbf{W}$	A trainable parameter matrix initialized with random values.
$\mathcal{T}(\cdot)$	A template to map the instance for prompt tuning.
$\mathbf{V}$	A label words set.
ϕ	A mapping function (also named as verbalizer).
$\mathcal{L}_{con}$	The contrastive loss.
$\mathcal{L}_{rp}$	The relation prediction loss.
$\mathcal{L}_{FSCL}$	The final loss consists of contrastive loss and relation prediction loss.
$d(\cdot, \cdot)$	Euclidean distance function.

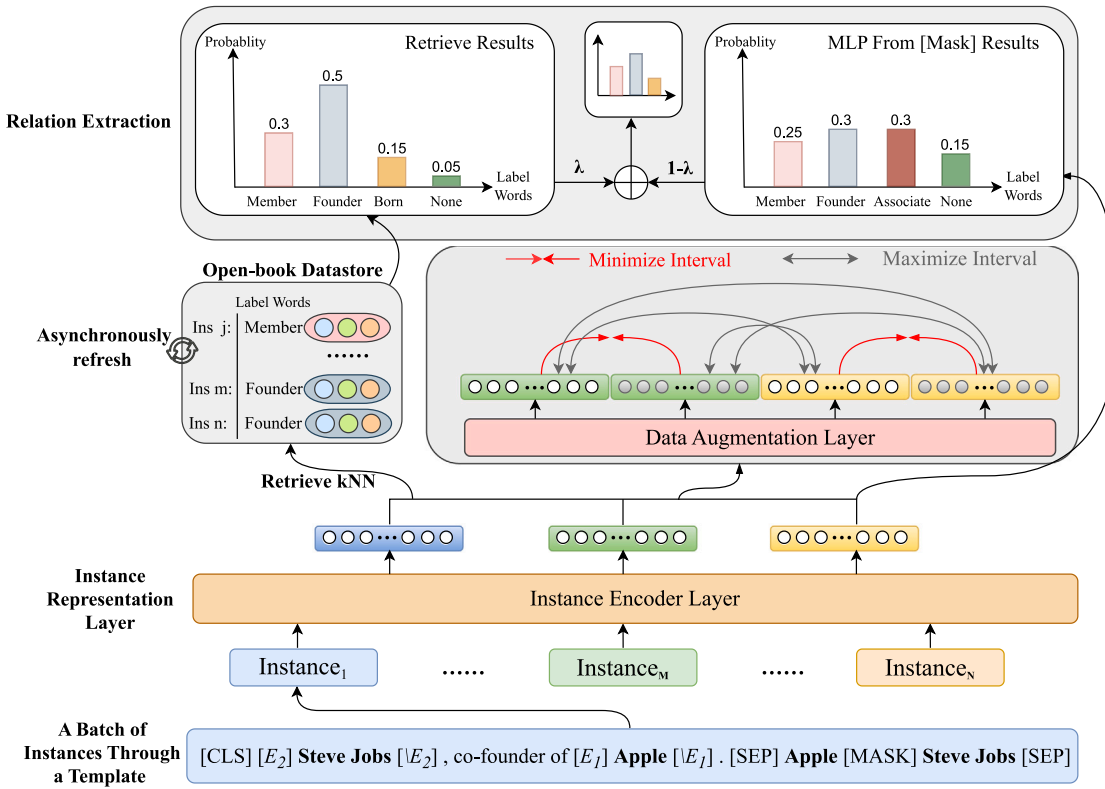


Fig. 3. Architecture of FSCL. It consists of four components and the details are introduced in Section 4.

At the end of each training epoch, the knowledge inside the open-book datasore will be updated asynchronously due to the fine-tuning of the input instance encoder, specifically, the sentences in the open-book datasore are re-inputted into the input instance encoder to obtain h_x , and the knowledge pairs with the class labels are reconstituted to refresh the open-book datasore. Step 4. By optimizing the contrastive loss function and classification loss function of relations, FSCL can make instances get a more distinctive representation in the representation space, with more subtle differences between instances. Step 5. Joint prediction is performed by the retrieval function in the open-book datasore and the well-optimized model in Step 4 to produce the final predicted probability of the relation.

4.2. Input instance encoder

Prompt-based instance representation is used to obtain instance embeddings after being sent to the instance encoder. Specifically, given

an instance x^i , a template function $\mathcal{T}(\cdot)$ is applied in x^i with entity pair subject and object to map x^i as $\hat{x}^i = \mathcal{T}(x^i)$, where $\hat{x}^i$ contains four special tokens such as $[\text{CLS}]$, $[\text{SEP}]$, $[\text{MASK}]$ and $[\text{E}]$. The input token instance $\hat{x}$ for input instance encoder is as follows:

$$\hat{x}^i = \{[\text{CLS}], x_1^i, \dots, x_j^i, [\text{E}_1], x_{sub}^i, [/\text{E}_1], \dots, [\text{E}_2], x_{obj}^i, [/\text{E}_2], \dots, [\text{SEP}], x_o^i, \dots, x_p^i, [\text{MASK}], x_r^i, \dots, x_L^i, [\text{SEP}]\}. \quad (1)$$

where $[\text{E}_1]$, $[/\text{E}_1]$, $[\text{E}_2]$, $[/\text{E}_2]$ are the beginning and end mark of the subject and object, respectively. More detailed information about the format and details of the samples in the dataset can be found in Appendix A. L is the length of the original instance. $[\text{CLS}]$ and $[\text{SEP}]$ are the special token and separator token, respectively. $[\text{MASK}]$, as well as the masked position token, is used to produce the hidden embedding $h_{[\text{MASK}]}$ for instance representation h_x . Then we employ PLMs to encode

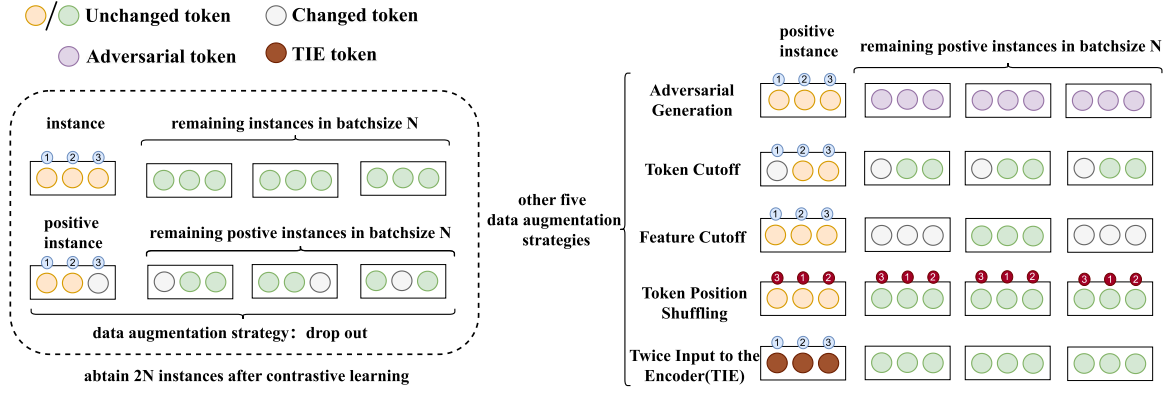


Fig. 4. The six data augmentation strategies we employ in this paper. During training with a batch size of N , we obtain $2N$ instances for contrastive learning through a data augmentation strategy.

and generate instance embeddings $h_i \in \mathbb{R}^{L \times d}$.

$$h_{\hat{x}^i} = \{h_{[\text{CLS}]}, h_{x_1^i}, \dots, h_{x_j^i}, h_{[E_1]}, h_{x_{sub}^i}, h_{[E_1]}, \dots, h_{[E_2]}, h_{x_{obj}^i}, h_{[E_2]}, \dots, h_{[\text{SEP}]}, h_{x_o^i}, \dots, h_{x_p^i}, h_{[\text{MASK}]}, h_{x_r^i}, \dots, h_{x_L^i}, h_{[\text{SEP}]} \}. \quad (2)$$

where d is the hidden state dimensions.

4.3. Contrastive learning module

Contrastive Instance Learning. In this section, a contrastive learning module is presented in order to learn more effective semantic differences and subtle differences between instances. For each input $h_{\hat{x}^i}$ from a batch of instances through a template using instance encoder, a transformation $T(\cdot)$ is applied to generate its augmented instance representation $\hat{h}_{\hat{x}^i}$, $\hat{h}_{\hat{x}^i} = T(h_{\hat{x}^i})$, where $\hat{h}_{\hat{x}} \in \mathbb{R}^{L \times d}$. We utilize $h_{[\text{MASK}]}$ as $h_{\hat{x}^i}$ and $\hat{h}_{[\text{MASK}]}$ as $\hat{h}_{\hat{x}^i}$ to construct an instance pair $\langle r_i, \hat{r}_i \rangle = \langle h_{[\text{MASK}]}, \hat{h}_{[\text{MASK}]} \rangle$.

We adopt the normalized temperature-scaled cross-entropy loss in-batch negative as the contrastive objective, which contributes to better instance representation. Specifically, We select N instances from the training set during every training step and construct a batch-set $\mathbf{S} = \{r_i, \hat{r}_i, r_j, \hat{r}_j, \dots, r_N, \hat{r}_N\}$ after augmentation. We regard $\langle r_i, \hat{r}_i \rangle$ as positive instance pair and other pairs as negative instance pairs:

$$\mathcal{L}_{i,\hat{i}} = -\log \frac{\exp(\text{sim}(r_i, \hat{r}_i)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\text{sim}(r_i, \hat{r}_k)/\tau)} \quad (3)$$

$$\mathcal{L}_{con} = \frac{1}{2N} \sum_{i=1}^{2N} \mathcal{L}_{i,\hat{i}} \quad (4)$$

where $r_i, \hat{r}_i \in \mathbf{S}$, $\text{sim}(\cdot)$ indicates the cosine similarity function, τ is a temperature hyperparameter, $\mathbb{1}$ is the indicator. Finally, we calculate the final contrastive loss $\mathcal{L}_{con}$ by averaging all $2N$ in-batch classification losses.

Data Augmentation Strategies. To learn finer-grained semantic representations between similar instances and semantic differences between different instances, we explore five different data augmentation strategies to generate instances for contrastive learning, including dropout [32], adversarial attack [33], cutoff [34], token shuffling and SimCSE [35]. More details are shown in Fig. 4.

- Dropout is an effective augmentation strategy and is a minimal form of data augmentation. In our experiments, We drop elements in instance embeddings at random with a particular probability and set their values to zero.
- Adversarial attack carefully crafted and often imperceptible perturbations are added to input text or image, with the intention of deceiving a model's predictions. We implement this strategy by following [36], which uses Fast Gradient Value (FGV) method to get perturbations.

- Cutoff [34] is proved to be a simple and efficient data augmentation that consists of three cutoff forms, token cutoff, feature cutoff, and span cutoff. In our experiments, the instance embeddings are created by the input instance encoder, and we randomly remove some token and feature dimensions (i.e. token and feature cutoff).
- Token shuffling initially aims to randomly shuffle the order of the tokens for the input instance. Inspired by this idea, we regard our instance representation r_i as a sequence that equipped orders. Thus, We add an additional position ID to each instance representation and then disrupt the position ID order when sending it to the encoder layer.
- SimCSE [35] adopts a simple data augmentation to generate its augmented view. Specifically, given an independent instance dropout mask, we simply feed the same input to the encoder twice (TIE) and get two embeddings as an instance pair. Note that this strategy is different from dropout since dropout drops a random token and sets it to zero.

4.4. Open-book datastore

We construct the open-book datastore from the training set by utilizing the contextual embedding and relation label. Specifically, given the i th instance $(I^i, \hat{y}^i)$ in the training dataset $\mathcal{D}$, we compute the representation-label pair $(h_{I^i}, \hat{y}^i)$, where h_{I^i} is the contextual embedding for instance I^i , and $\hat{y}^i$ is the relation class label, respectively. We construct the open-book datastore $(\mathcal{Q}, \mathcal{V})$ with storing all pairs $(h_{I^i}, \hat{y}^i)$ and denote h_{I^i} as *query* and $\hat{y}^i$ as *value* as follows:

$$(\mathcal{Q}, \mathcal{V}) = \{(h_{I^i}, \hat{y}^i) \mid (I^i, \hat{y}^i) \in \mathcal{D}\} \quad (5)$$

Our encoder layer may result in a variable contextual embedding of the instance as the PLM parameters are continuously updated, so we propose to “refresh” the open-book datastore by asynchronously re-embedding all instances every T training epochs. Taking different values of the asynchronous refresh steps may affect the final result of retrieval, hence, we experimented with three parameters, using 1, 2, and 4 as the training epochs on all the datasets, and the final result was that 1 gave the best results, and four of the five datasets were sub-optimal when the training epochs was 4.

4.5. Relation prediction

In relational prediction, we make predictions by synthesizing the output from the instance encoding layer and retrieving the open-book datastore. Specifically, We utilize h_{r_i} from the instance encoder, and

Table 2

More details about datasets, including the number of training, validation, and test set, also containing the number of relation types.

Dataset	# Train.	# Val.	# Test.	# Rel.
SemEval	6,507	1,493	2,717	19
DialogRE	5,963	1,928	1,858	36
TACRED	68,124	22,631	15,509	42
TACREV	68,124	22,631	15,509	42
Re-TACRED	58,465	19,584	13,418	40

the probability distribution over training dataset D can be formalized by Multi-layer Perceptron(MLP) as follows,

$$P(y^i | x^i, \theta) = \text{softmax}(W_h(\text{sigmoid}(h_{r_i})) + b_h) \quad (6)$$

where x^i and y^i are the i th instance and corresponding relation label during framework learns. θ is the model parameter, $W_h \in \mathbb{R}^d$ and $b_h \in \mathbb{R}^d$ are learnable parameters. For each instance input x_i , we employ cross-entropy to calculate relation prediction loss.

$$\mathcal{L}_{rp} = \frac{1}{N} \sum_{i=1}^N \text{CrossEntropy}(P(y^i | x^i, \theta), \hat{y}^i) \quad (7)$$

where $\hat{y}^i$ is the ground-truth label of the i th instance. Then, we incorporate retrieval into relation prediction by querying the open-book datastore $(Q, \mathcal{V})$ to retrieve the k -nearest neighbors $\mathcal{N}$ of h_{x^i} , according to a distance function $d(\cdot, \cdot)$, where $d(\cdot, \cdot)$ typically adopt the Euclidean distance. To form the probability distribution over pre-defined relations, to compute the distribution over neighbors and aggregate probability mass, we employ the softmax function over negative distances.

$$P_{kNN}(y^i | \mathcal{T}(x^i)) \propto \sum_{(h_{r_i}, \hat{y}^i) \in \mathcal{N}} 1_{y=\hat{y}^i} \exp(-d(h_{r_i}, h_{x^i})). \quad (8)$$

Lastly, we reformulate relation prediction by interpolating the non-parametric k nearest neighbor distribution P_{kNN} with Eq. (6) $P(y_i | x_i, \theta)$ using parameter λ to produce the final probability of the relation:

$$P(y^i | x^i, \theta) = \lambda P_{kNN}(r | \mathcal{T}(x^i)) + (1 - \lambda) \text{softmax}(W_h(\text{sigmoid}(h_{r_i})) + b_h). \quad (9)$$

In the training phase, we use the sum of the two loss functions as the training objective of the model. The contrastive loss is mainly used to enhance the differentiation between instances of different classes by pulling instances of the same class closer together and pushing instances of different classes further apart, thereby learning richer and more distinctive feature representations. The classification loss, on the other hand, focuses on improving the model's predictive accuracy for relation classes, ensuring that the model can accurately identify relations even under conditions of limited samples. The training objective of FSCL is as follows,

$$\mathcal{L}_{FSCL} = \alpha \mathcal{L}_{con} + \mathcal{L}_{rp} \quad (10)$$

where α is a hyper-parameter to balance two loss functions.

5. Experiments

5.1. Datasets

Five datasets are used to evaluate our FSCL model: SemEval 2010Task8(SemEval) [37], DialogRE [38], TACRED [39], TACRED-Revisit [40], Re-TACRED [41]. More details about datasets² are shown in Table 2.

² Datasets are available in https://drive.google.com/file/d/1BQW3ekxJ96w2652FndRtcEIHp6Gwy_py/view?usp=sharing or https://github.com/JohnnyGWY/FSCL_Dataset.

5.2. Implementation details

We deploy Roberta-large as model initialization for all datasets while deploying Roberta-base for DialogRE to get a fair experiment setting compared with baselines. More detailed settings can be found in Appendix B.

Standard Setting. We compare with the following methods to evaluate our proposed FSCL in standard setting: (1) FINETUING adopts a standard fine-tuning approach for PLMs to relation extraction. (2) SPAN-BERT [18] is a span-level fine-tuning method that masks continuous spans and introduces span targets for relation extraction, (3) KNOW-BERT [19] enhances the representation of entities from the knowledge base and re-contextualizes them. (4) LUKE [20] utilizes entity-aware self-attention to contextualize embeddings of words and entities. (5) MTB [42] designs a novel pre-training task Matching the Blanks and solves the relation extraction task without manual labeling. (6) GDPNET [11] refines latent multi-view graph for relation extraction. (7) DUAL [43] constructs dialogue-level abstract meaning representation (AMR) graphs from sentence-level AMRs for dialogue modeling. (8) PTR [23] provides strategies for prompt-tuning RE by dividing relations into sub-prompts using logical rules. (9) KNOWPROMPT [17] incorporates virtual relation word among relation labels into prompt-tuning. (10) RETRIEVALRE [4] utilizes retrieval-augmented to enhance prompt-tuning for few-shot relation extraction. (11) KLG [44] constructs a top-k candidate prediction set of relation labels and utilizes graph networks to aggregate relation representations. (12) MVRE [22] decouples relational representations through multi-view graphs and optimizes the learning of relational representations.

Low-Resource Setting. We conduct 1-, 5- and 16-shot experiments following [4], specifically, we randomly select k instances of each relation label from the initial training and validation sets to create the few-shot training and validation sets. We repeat five times with different random seeds and report the average result and variance. We also realize that selecting different pre-trained language models as input instance encoders may further affect the performance results. In order to explore the dependence of FSCL on the pre-trained models and provide a reference point for future research, we select DeBERTa Xlarge [45] and DeBERTa XXlarge [45] to conduct experiments on TACRED under low resource settings, and the specific results can be found in Appendix C.

Evaluation Metrics. For evaluation metrics, micro F_1 (Eqs. (11)–(13)) is adopted as our primary metrics, considering that F_1 evaluates the overall performance of precision and recall. In the standard setting, where some certain relations will have a limited number of instances available, micro F_1 metrics can treat all relations equally, overcome label imbalances, and synthesize overall performance. In low-resource setting, although the number of every relation label is certain and identical, using micro F_1 metrics contributes to the global performance, that is, the combined performance of the model across all relation types.

$$\text{micro } F_1 = \frac{2 \times (\text{micro precision} \times \text{micro recall})}{\text{micro precision} + \text{micro recall}} \quad (11)$$

$$\text{micro precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}} \quad (12)$$

$$\text{micro recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}} \quad (13)$$

5.3. Experimental results

Table 3 shows the main results, where FSCL outperforms the other baselines in terms of Micro F_1 values on the four different datasets in a standard supervised setting. Specifically, on the TACRED dataset, FSCL's performance ranks just below KLG. This is attributed to TACRED represents a classic case of long-tail distribution imbalance. Aside from the “no relation” class, the sum of frequencies of the top 10 most

Table 3

RE with standard setting performance of Micro- F_1 scores (%) on different datasets. Results are all cited from published papers or re-implemented using official code. Note that we exceptionally run the PTR, KnowPrompt, RetrievalRE, KLG and MVRE with Roberta-base in DialogRE to ensure a fair comparison. The best results are marked in bold.

Standard Supervised Setting					
Methods	SemEval	DialogRE	TACRED	TACREV	Re-TACRED
Fine-tuning pre-trained models					
FINE-TUNING	87.6	57.3	68.7	76.0	84.9
SPANBERT [18]	–	–	70.8	78.0	85.3
KNOWBERT [19]	89.1	–	71.5	79.3	89.1
LUKE [20]	–	–	72.7	80.6	–
MTB [42]	89.5	–	70.1	–	–
GDPNET [11]	–	64.9	71.5	79.3	–
DUAL [43]	–	67.3	–	–	–
Prompt-tuning pre-trained models					
PTR [23]	89.9	63.2	72.4	81.4	90.9
KNOWPROMPT [17]	90.2	68.6	72.4	82.4	91.3
RETRIEVALRE [4]	90.4	69.4	72.7	82.7	91.5
KLG [44]	90.7	69.5	76.0	82.5	84.8
MVRE [22]	90.7	69.8	72.8	83.1	91.3
FSCL (OURS)	91.3	70.1	73.3	83.5	91.9

Table 4

RE with low-resource setting performance of micro- F_1 scores (%) on three public datasets. We repeat five times with different randomly seeds and report the average result and variance. The best results are highlighted.

Dataset	Low-Resource Setting								
	SemEval			TACRED			TACREV		
Model	K = 1	K = 5	K = 16	K = 1	K = 5	K = 16	K = 1	K = 5	K = 16
FINE-TUNING	18.5 (± 1.4)	41.5 (± 2.3)	66.1 (± 0.4)	7.6 (± 3.0)	16.6 (± 2.1)	26.8 (± 1.8)	7.2 (± 1.4)	16.3 (± 2.1)	25.8 (± 1.2)
GDPNET	10.3 (± 2.5)	42.7 (± 2.0)	67.5 (± 0.8)	4.2 (± 3.8)	15.5 (± 2.3)	28.0 (± 1.8)	5.1 (± 2.4)	17.8 (± 2.4)	26.4 (± 1.2)
PTR	14.7 (± 1.1)	53.9 (± 1.9)	80.6 (± 1.2)	8.6 (± 2.5)	24.9 (± 3.1)	30.7 (± 2.0)	9.4 (± 0.7)	26.9 (± 1.5)	31.4 (± 0.3)
KnowPrompt	28.6 (± 6.2)	66.1 (± 8.6)	80.9 (± 1.6)	17.6 (± 1.8)	28.8 (± 2.0)	34.7 (± 1.8)	17.8 (± 2.2)	30.4 (± 0.5)	33.2 (± 1.4)
RetrievalRE	33.3 (± 1.6)	69.7 (± 1.7)	81.8 (± 1.0)	19.5 (± 1.5)	30.7 (± 1.7)	36.1 (± 1.2)	18.7 (± 1.8)	30.6 (± 0.2)	35.3 (± 0.3)
MVRE	54.6 (± 2.8)	77.6 (± 3.6)	82.5 (± 0.8)	21.2 (± 2.2)	32.4 (± 1.2)	34.8 (± 0.8)	20.5 (± 1.9)	31.0 (± 1.4)	34.3 (± 1.1)
Ours (Adversarial attack)	34.0 (± 0.9)	69.9 (± 1.3)	82.3 (± 1.7)	20.4 (± 1.5)	31.2 (± 1.4)	36.6 (± 1.0)	19.6 (± 2.0)	31.1 (± 0.3)	36.1 (± 0.4)
Ours (Feature cutoff)	33.7 (± 1.1)	70.2 (± 2.0)	82.1 (± 1.8)	20.3 (± 1.2)	31.2 (± 1.6)	36.9 (± 1.3)	19.9 (± 1.3)	30.8 (± 0.6)	35.7 (± 0.4)
Ours (Token cutoff)	36.4 (± 1.1)	72.5 (± 1.7)	83.0 (± 1.4)	23.4 (± 1.6)	32.7 (± 1.6)	37.7 (± 0.9)	22.0 (± 1.7)	31.9 (± 1.1)	36.8 (± 1.5)
Ours (Token shuffling)	35.1 (± 2.6)	71.7 (± 3.0)	82.4 (± 1.1)	22.7 (± 1.4)	31.8 (± 1.2)	37.0 (± 1.3)	20.5 (± 1.3)	31.7 (± 0.9)	37.5 (± 0.6)
Ours (SimCSE)	35.3 (± 1.9)	71.7 (± 2.3)	82.6 (± 1.8)	23.0 (± 2.1)	31.7 (± 1.5)	37.4 (± 2.2)	20.4 (± 2.1)	31.4 (± 1.8)	39.1 (± 1.8)
Ours (Dropout)	36.7 (± 1.4)	72.1 (± 1.9)	83.1 (± 1.0)	23.9 (± 0.9)	33.0 (± 1.7)	38.1 (± 0.9)	21.7 (± 1.6)	32.7 (± 0.4)	37.8 (± 0.6)

common classes accounts for 69.6%, while the sum of frequencies of the bottom 10 categories is a mere 3.6%, which is precisely the initial motivation for the KLG. However, our approach has achieved improvements over the KLG method across the other four datasets. It can be noted that LUKE, KnowBERT, SpanBERT, and KnowPrompt introduce knowledge as data augmentation to obtain a better representation of the relation. In contrast, we focus on the data itself and reduce the introduction of knowledge by constructing a positive pair of instances in a batch size to unite the remaining negative pairs for training, allowing instances in the same class to be clustered together, and the instances of different classes are separated, which bolsters the model's capability to retain robust performance when encountering new and unseen data. FSCL also gains improvements over RetrievalRE, although RetrievalRE utilizes retrieval to mitigate the deficiency or instability of specific relational knowledge learned in the pre-trained model rather than using data augmentation, the instances on which retrieval is constructed are not themselves well characterized, and FSCL learned more discriminatory features.

As shown in Table 4, we can observe that FSCL improves on all baselines, and when the training data is unusually sparse, FSCL achieves an increase in Micro F_1 scores that not only improves but also has a comparable variance to the current state-of-the-art models. This inspires us that in the case of sparse few-shots, instances with rich information and strong representations are required, because instances with similar representations are not critical for learning the weights of a large language model, and that updating the open-book datastore in real-time during retrieval leads to improved performance. It is worth noting that when k is set to 1 or 5, FSCL does not perform as well as MVRE in SemEval, but it performs better than MVRE in TACRED and TACREV. We attribute this to two main reasons: (1) MVRE decouples each relation into different perspectives to construct a multi-view relational representation. MVRE is more advantageous when k is small and the semantics of relation labels are discriminative, e.g., “Cause-Effect” and “Member-Collection” in SemEval. (2) TACRED

and TACREV have more similar relations compared to SemEval, such as “org:city_of_headquarters” and “org:stateorprovince_of_headquarters”. In low-resource scenarios, FSCL can learn to distinguish features of similar relations, and the improvement becomes more pronounced when k is set to 16. Not only does FSCL performs well with the data augmentations, but also the results demonstrate the robustness of our approach in the data scarcity scenarios. Dropout and token cutoff are the two most effective strategies, where dropout is slightly better than token cutoff and outperforms SimCSE and token shuffling. This may be due to the fact that dropout and token cutoff alter instance representations more strongly in the presence of sparse few-shot examples, making them more relevant to downstream RE tasks.

5.4. Qualitative analysis

Effect of α in loss function and λ in relation prediction. Since the loss function of FSCL is composed of two parts, relation prediction and contrastive learning, α is introduced to balance the weight of the two parts, we analyze and study the effect of the change of α on the results. As shown in Fig. 5, the model achieves the best results on 16-shot SemEval dataset when α is set to 0.001. At this point, the scales of the two parts are balanced, for the learning of the loss function to be effective. Under this setting, the contrastive loss effectively complements the classification loss, enabling the model to better distinguish different types of relations even with a limited amount of labeled data. Therefore, we believe that the contrastive loss serves an auxiliary and enhancing role within the model, providing an optimized representational space for the classification loss, thereby jointly promoting the improvement of the FSCL's performance. Similarly, we add λ to the final relation prediction to control the weight of retrieval and thus correct for erroneous predictions. At first we thought that the introduction of the retrieval module part with the asynchronous refresh is an enhancement, that is, the function of auxiliary prediction, the value of λ is floating in a small range interval, so we set λ to 0.1 firstly,

with the increase of λ the effect of the model is as illustrated in Fig. 5, when λ is set to 0.2, the model works best.

Effect of dropout ratio and temperature hyperparameter τ . Since the temperature hyperparameter τ in the contrastive learning module affects the optimization of the loss function and the model performance in the future. The smaller the temperature coefficient, the more attention is paid to the hard negative instances that are very similar to that positive instance, giving the hard negative instances a larger gradient to separate them from the positive instance, and overforcing the separation from the hard negative instances destroys the underlying semantic structure learned. As the work in [46] argues, a good temperature coefficient should be a compromise between uniformity and tolerance. Thus, following previous experience, we set a very small interval for the temperature hyperparameters, within which we study the parameter values suitable for the model. As shown in Fig. 6, all the results have a large variance, but the model works best when the temperature hyperparameter is set to 0.065. We further explore the impact of different dropout rates on model performance which has proved to be the best performance data augmentation strategy in previous our experiments. As shown in Fig. 6, when the dropout rate is set to 0.3, not only better performance but also lower variance is achieved on the RETACRED dataset.

Effect of batch size N in training and k -nearest neighbors in retrieval. The batch size is a pivotal hyperparameter in the training process, significantly influencing the training dynamics and the ultimate performance of the model. In our experiments, we explore various batch sizes to understand their impact on FSCL. As shown in Fig. 7, we experiment with batches of 8, 16, 32, and 64. Smaller batch sizes tend to provide more regular updates to the model's weights, which could lead to a higher frequency of weight updates in FSCL, resulting in a more refined and stable convergence of the learning process, albeit at a slower training speed. On the other hand, larger batch sizes can accelerate the training process but may result in less stable updates. Our empirical results indicate that a batch size of 8 achieves a good balance between training speed and stability. In the process of constructing the open-book datastore for retrieval, the size of the k -nearest neighbors (k -NN) may affect the model's learning process. The k -NN relies on the quality of instance representations; if the instances are not sufficiently distinctive, the k -NN retrieval may include instances that are not truly similar, which could impact FSCL's predictive outcomes. As shown in Fig. 7, FSCL performs best when k is set to 16, and as the value of k increases, the performance gradually reaches a threshold. This demonstrates that the contrastive learning in FSCL can enhance the distinctiveness of instance representations, stabilizing the retrieval process and thereby supporting in the enhancement of the final relation prediction.

5.5. Ablation study

We conduct an ablation study on each module by correspondingly ablating one in order to more thoroughly validate our model, and the results are shown in Table 5. For-Asynchronous Refresh, we remove asynchronous refresh in the open-book datastore. For the-Retrieval Module, we ablate the retrieval module from our model. In many natural language task, [CLS] is also used as the classification token and gain well performance, so For-Relation Prediction ([CLS]), We replace the representation of [MASK] in the MLP relation prediction with the representation of [CLS] token for training and prediction, and for-Contrastive Learning Module, we remove the contrastive learning module.

From Table 5, the experimental results demonstrate that our proposed module is indispensable for model performance improvement. After removing the contrastive learning module, the model performance drops significantly, especially in the low-resource setting, with decreases of 2.0, 1.7, and 1.6, respectively, which also demonstrates that it is very crucial to obtain an instance representation with rich

Table 5

Ablation study on RE-TACRED.

Model	K = 1	K = 5	K = 16	Full
FSCL	25.0	51.3	67.1	91.9
-Asynchronous Refresh	24.3	50.8	66.3	91.5
-Retrieval Module	23.5	50.7	66.2	91.3
-Relation Prediction ([CLS])	23.8	50.7	66.2	91.0
-Contrastive Learning Module	23.0	49.6	65.5	90.7

information for relation extraction in the low-resource environment. The utilize of the contrastive learning module cannot only form a reliable open-book for retrieval assistance in the retrieval part but also further learn the differences between instance relations in the relation prediction ([MASK]). Although the impact of contrastive learning becomes weaker as the number of instances increases, this still proves that the inclusion of a contrastive learning module is effective.

We also find that our retrieval module and relation prediction ([MASK]) are complementary. The retrieval module further classifies the instance representations obtained from contrastive learning, focuses on summarizing the knowledge between semantically similar instances, and implicitly learns the knowledge of the differences between relations. The relation prediction module, on the other hand, can use the relations of instance representations after contrastive learning to further explicitly learn the differences between relations, but cannot summarize the knowledge between semantically similar instances. Removing any of the modules does not fully utilize the ability of contrastive learning, which in turn leads to a decrease in model performance. Thus these modules organically form our model through the above connections.

5.6. Complexity analysis

In this section, we analyze the computational complexity of our proposed FSCL, particularly focusing on the contrastive learning method employed in few-shot relation extraction tasks. **Contrastive Learning Complexity:** The primary source of computational complexity in our framework stems from the contrastive learning process, which is designed to improve the model's generalization ability by distinguishing feature representations of different samples. The naive implementation of contrastive learning can have a complexity of $O(N^2)$, especially when dealing with large datasets where N represents the number of samples. This complexity arises from the need to compare each sample with every other sample in the dataset. **Optimization Strategies:** To mitigate this complexity, we have implemented several optimization strategies: **1. Negative Sampling:** Instead of comparing a sample with all other samples, we employ a negative sampling technique. Negative samples are constructed from all samples in a batch, excluding the original one. After data augmentation, this approach reduces the computational complexity from $O(N^2)$ to $O(k \times N)$, where k is the number of negative samples in a batch. Typically, k is much smaller than N , significantly lowering the computational burden. **2. Mini-Batch Processing:** We further optimize computational efficiency by utilizing mini-batch processing, where multiple samples are processed in parallel during each iteration. This method not only improves computational efficiency but also leverages modern hardware's parallel computing capabilities. We experimented with mini-batch sizes of 16 and 32, finding that a batch size of 16 generally performs better across multiple datasets. **3. Asynchronous Update of Open-Book Datastore:** An additional optimization is the asynchronous update of the open-book datastore during training. This method saves time during final predictions and further enhances performance. Consequently, the average inference time for our few-shot experiments is reduced to between 1 and 1.5 h. **Comparison with Other Methods:** The baselines referenced in this paper also utilize different strategies to reduce computational

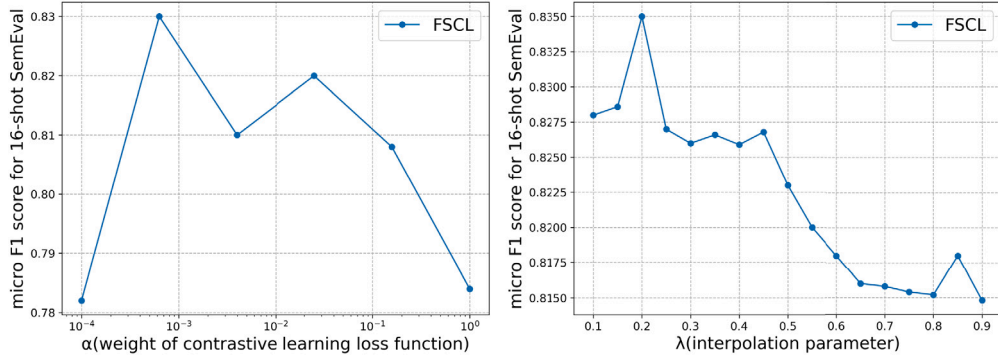


Fig. 5. The effect of the variation of the two hyperparameters α and λ , where the results are obtained on the 16-shot SemEval dataset.

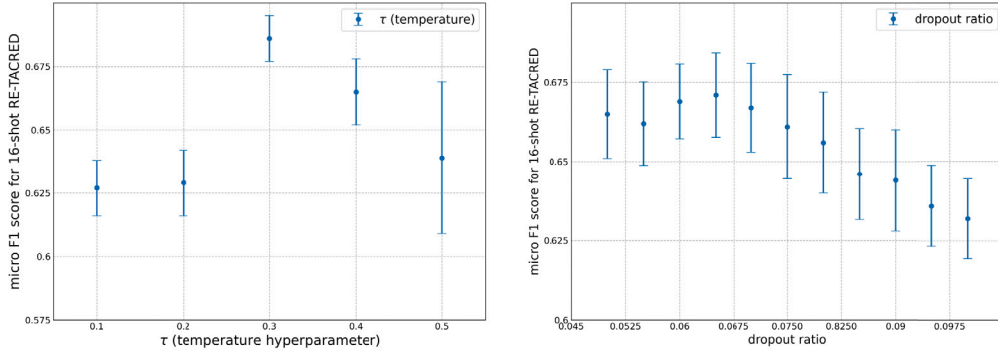


Fig. 6. The effect of the variation of ratio in dropout and temperature hyperparameter τ . The results are obtained on the 16-shot Re-TACRED dataset. Note that we select five random seeds for our experiments and obtained the average performance and its variance.

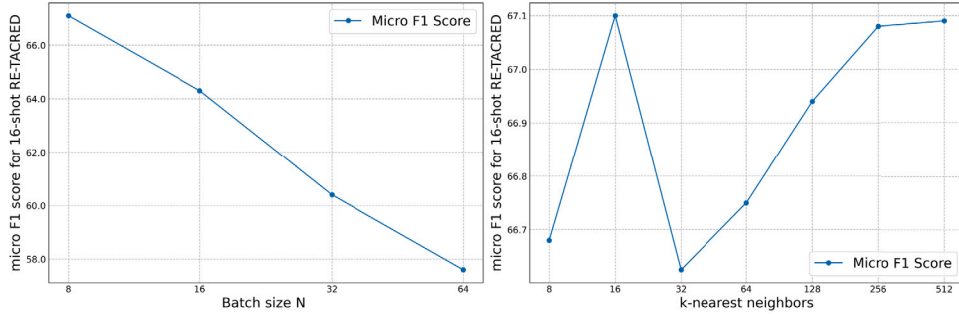


Fig. 7. The effect of batch size N in training and k -nearest neighbors in retrieval. We conduct 16-shot experiments on Re-TACRED dataset.

complexity, each achieving a complexity lower than $O(N^2)$. These methods, while varied in their approaches, share the common goal of improving efficiency and scalability in few-shot relation extraction tasks.

6. Case study

We argue that although previous reasoning methods for relation extraction are able to handle the scarcity of relations on a small number of instances, they cannot represent semantics effectively and thus the knowledge of similar instances and similar relations will not be gained sufficiently.

In this section we conduct a case study to illustrate how our model changes the distribution of relation predictions. Specifically, we extract two instances with high semantic similarity on the SemEval dataset, where the gold relation for instance 1 is *Member-Collection*, and the gold relation for instance 2 is *Component-Whole*, respectively. The previous relation extraction inference approach consists of two parts, the first

part is done by MLP prediction, which is done using the values of masked token [MASK] after the instance has been encoded by the encoder. The second part is the augmented prediction aided by the retrieval module. As can be seen from Fig. 8, the previous predictions which use MLP predictions for the first part of the two instances are the same, and the retrieval module's retrieval results lead to the first instance's relation prediction to be the fourth: *Component-Whole*, which is an incorrect relation; whereas the prediction result of the second instance is consistent with the gold relation, and the prediction is correct.

The prediction of our model is likewise composed of two parts, the first part is the [MASK] prediction, and the second part is likewise composed of the retrieval module. As shown in our prediction in Fig. 8, after contrastive learning, the first part distribution of relations of instance 1 and instance 2 changes, the distribution of instance 1 on relation 3 and 4 changes from being equal to relation 3 being dominant, and the probability of the distribution of relation 3 is further enhanced in the subsequent retrieval module (i.e. the second part),

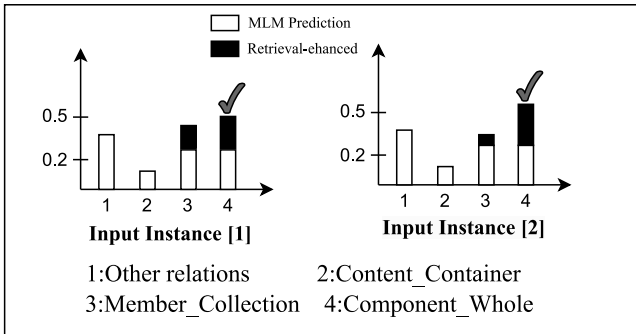
Input Instance [1] : A **fling**(e_1) of **oxbirds**(e_2) is known by the singular and beautiful appearance those birds present in the air; the grace,swiftness, and compactness of their flight.

Gold relation(e_2, e_1): Member-Collection

Input Instance [2] : **Birds**(e_1) with eyes on the sides of their heads have a wide visual field,while birds with **eyes**(e_2) on the front of their heads,such as owls,have binocular vision.

Gold relation(e_2, e_1): Component-Whole

Previous Prediction



Our Prediction

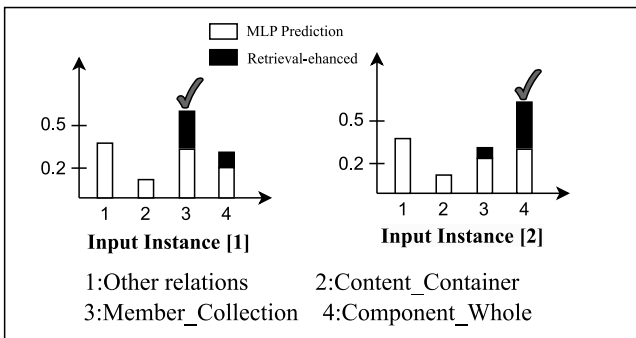


Fig. 8. The case study we conduct. “previous prediction” means that we utilize the same baseline method as Fig. 1 as a prediction. Note that we sum the relations with low predictive probability and label them as “Other relations”.

finally the relation is predicted correctly. The same is true for instance 2. This demonstrates how the instance representations altered during contrastive learning, allowing the MLP to learn variations between relations. and the semantics between similar instances to be further learned by the retrieval module.

7. Conclusions

In this paper, we propose a novel framework with few-shot Instance-wise contrastive learning and open-book datastore for relation extraction to address the similar instances and similar relations in an instance-scarce environment. In conclusion, our study leverages augmented pairs during the training phase and implements asynchronous refresh on the open-book datastore to bolster the effectiveness of the retrieval module in relation extraction. The proposed framework is validated on five public datasets, and the comprehensive evaluation and analysis show its state-of-the-art performance. The ablation experiments and case studies are also conducted and further demonstrate the effectiveness of the proposed framework. Our augmented pairs strategy contributes to the model’s robustness, while the asynchronous refresh

mechanism plays a pivotal role in enhancing the retrieval module’s efficiency. The case studies further illustrate the real-world applicability and impact of our framework in diverse scenarios.

8. Limitation and future work

Although our FSCL utilizes contrastive learning to obtain a more informative representation of the instances and achieves good results in FSRE, it still has limitation that can be addressed. The quality of these instance representations is heavily dependent on the selection of negative instances. If there is an overlap in the representations of augmented views from instances of different relation classes after data augmentation, it could significantly degrade model performance. Even though we have an open-book datastore to assist in the prediction, but the open-book datastore is also constructed based on the instance representations. Therefore, our future research aims to generate more distinctive representations for each instance to address potential overlaps in augmented views, enhancing the robustness of model. This endeavor may involve exploring more advanced preprocessing techniques or leveraging additional contextual information to strengthen the understanding of instances and relations in limited data scenarios.

CRedit authorship contribution statement

Wanyuan Gong: Writing – original draft, Software, Methodology, Conceptualization. **Qifeng Zhou:** Writing – review & editing, Validation, Methodology, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Instance details

In this section, we show the format and details of the datasets used in FSCL:

Instance 1:{"token": ["he", "has", "always", "waited", "patiently", "in", "the", "studio", "until", "his", "master", "s", "voice", "signals", "that", "it", "was", "time", "to", "go", "home", "."], "h": {"name": "master", "pos": [10, 11]}, "t": {"name": "voice", "pos": [12, 13]}, "relation": "Component-Whole(e_2, e_1)"}

Instance 2:{"token": ["His", "death", "was", "confirmed", "by", "Rick", "Doblin", ",", "the", "Boston-based", "founder", "of", "the", "Multidisciplinary", "Association", "for", "Psychedelic", "Studies", ",", "a", "nonprofit", "pharmaceutical", "company", "developing", "LSD", "and", "other", "psychedelics", "for", "prescription", "medicines", "."], "h": {"name": "Multidisciplinary Association for Psychedelic Studies", "pos": [13, 18]}, "t": {"name": "Rick Doblin", "pos": [5, 7]}, "relation": "org:founded_by"}

There are five key tokens that need to be explained:

"token": the tokenized processed sentence.

"h": name and position of the head entity in the instance.

"t": the name and position of the tail entity in the instance.

"pos": the start and end position in instance for entity.

"relation": the relation between two entities in the instance, the relation in the instance is “Product-Producer(e_2, e_1)”, which means that entity 1 (the head entity) is Producer and entity 2 (the tail entity) is

Product. If relation is “org:founded_by”, it means the head entity is founded by the tail entity.

After processing with our template, we consider the subject and head entity as identical, and the object and tail entity as identical. We have used the terms “subject” and “object” and have applied [E1], [/E1], [E2], [/E2] as token markers for the start and end positions of the subject and object entities.

Appendix B. Hyper-parameters setting

This section details the training procedures and hyperparameters for each of the datasets. We utilize Pytorch to conduct experiments with one Nvidia 4090 GPU and Huggingface Transformers. All optimizations are performed with the AdamW optimizer. We set five numbers in the range 1–5 for the random seed. In our experiments we set a search space for certain hyper parameters, where the values of the hyper-parameters with the best predictions are bolded.

B.1. Standard setting

The hyperparameters search space of FSCL in standard setting is shown as Follows:

- K nearest neighbors: [8, **16**, 32, 64, 128]
- learning rate: [$2e - 5$, $3e - 5$, **$5e - 5$**]
- interpolation parameter: 0.2
- max instance length: [**256**, 512]
- batch size: 16 (for TACRED-Revisit, Re-TACRED and DialogRE as 8)
- temperature: [0.035, 0.05, **0.065**, 0.08]
- number of epochs: 16 (for DialogRE as 20)
- gradient accumulation steps: 1 (for DialogRE as 4)
- drop out ratio: 0.30
- token cutoff ratio: 0.15
- feature cutoff ratio: 0.2

B.2. Low-resource setting

The hyperparameters search space of FSCL in low-resource setting is shown as Follows:

- K nearest neighbors: [8, **16**, 32, 64, 128]
- learning rate: [$2e - 5$, $3e - 5$, **$5e - 5$**]
- interpolation parameter: 0.2
- max instance length: [**256**, 512]
- batch size: 16 (for TACRED-Revisit, Re-TACRED and DialogRE as 8)
- temperature: [0.035, 0.05, **0.065**, 0.08]
- number of epochs: 40 (for DialogRE as 32)
- gradient accumulation steps: 1 (for DialogRE as 4)
- drop out ratio: 0.30
- token cutoff ratio: 0.15
- feature cutoff ratio: 0.2

Appendix C. Dependence on different models

We present here the experimental results of FSCL’s dependency on pre-trained models, utilizing two models from the DeBERTa series, DeBERTa xlarge and DeBERTa xxlarge, for comparison. DeBERTa (Decoding-enhanced BERT with disentangled attention) is a pre-trained language model introduced by Microsoft Research that innovates upon the BERT framework with the addition of decoding enhancement and disentangled attention mechanisms. These improvements have endowed the DeBERTa series of models with superior performance across a variety of natural language processing tasks. The DeBERTa

Table C.6

Experimental results for different pre-trained models as input instance encoders. The table shows the experimental results of 1-, 5-, and 16-shot experiments on the TACRED dataset under low resource settings.

Instance encoder model	1-shot	5-shot	16-shot
RoBERTa-large	23.9	33.0	38.1
DeBERTa xlarge	25.5	34.7	40.7
DeBERTa xxlarge	27.7	36.7	41.4

xlarge model contains approximately 750 million parameters, while the DeBERTa xxLarge boasts 1.5 billion parameters. They both perform well on the GLUE benchmark and the SQuAD question–answer dataset, often achieving higher scores on these tasks than BERT Large and RoBERTa.

As illustrated in the Table C.6, our experimental results have improved after selecting the pre-trained model with superior overall performance. However, this improvement has come at the expense of increased inference time, which has risen from an average of 1 to 1.5 h to 3.5 to 4 h. This indicates that our method is dependent on the performance of the pre-trained model while also validating the scalability of FSCL’s ideas on the models. It is noteworthy that current baseline methods do not discuss the dependency on different pre-trained models.

Data availability

Data will be made available on request.

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